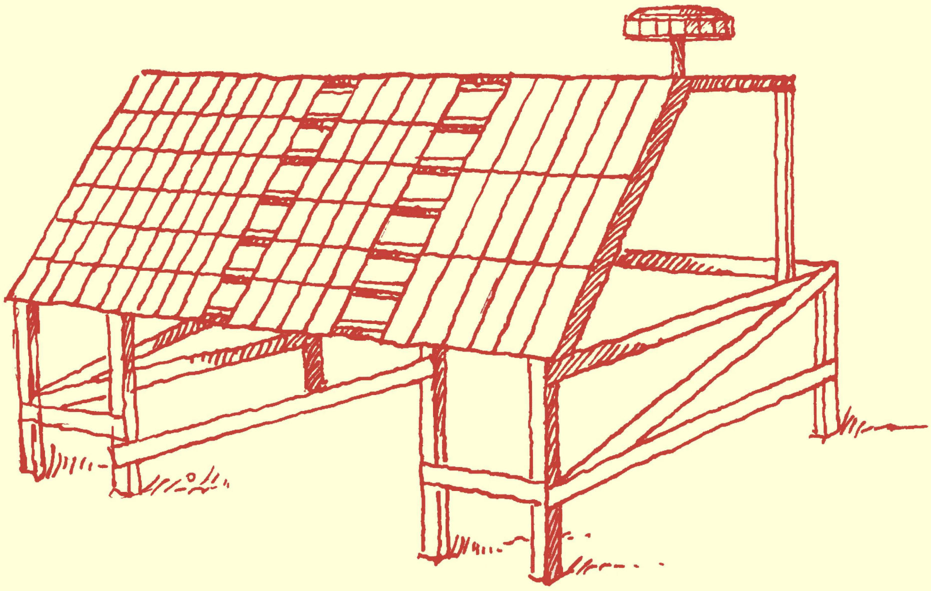
CONVENTION ON LONG-RANGE TRANSBOUNDARY AIR POLLUTION

UN/ECE INTERNATIONAL CO-OPERATIVE PROGRAMME ON EFFECTS ON MATERIALS, INCLUDING HISTORIC AND CULTURAL MONUMENTS



Report No 62:

Results of corrosion and soiling from the 2008–2009 exposure programme for trend analysis.

August 2010

PREPARED BY THE MAIN RESEARCH CENTRE

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ICP Materials
http://www.corr-institute.se/ICP-Materials

Report No 62
Results on corrosion and soiling from the 2008–2009
exposure programme for trend analysis.

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Contents

1 Introduction

Early in the discussions on the "Convention on Long-range Transboundary Air Pollution" it was recognized that a good understanding of the harmful effects of air pollution was a prerequisite for reaching agreement on effective pollution control. To develop the necessary international cooperation in the research on and the monitoring of pollutant effects, the Working Group on Effects (WGE) was established under the Convention in 1980 and held its first meeting in 1981. The Convention involves countries in the UNECE region and has its secretariat with the UNECE [1].

The Working Group on Effects provides information on the degree and geographic extent of the impacts on human health and the environment of major air pollutants, such as sulphur and nitrogen oxides, ozone and heavy metals. Its six International Cooperative Programmes (ICPs) and the Task Force on Health identify the most endangered areas, ecosystems and other receptors by considering damage to human health, terrestrial and aquatic ecosystems and materials. An important part of this work is long-term monitoring. The work is underpinned by scientific research on dose-response, critical loads and levels and damage evaluation [1].

The International Co-operative Programme on Effects on Materials including Historic and Cultural Monuments (ICP Materials) is one of the six ICPs. During the course of the programme, which had its first exposure in 1987, the special importance of trend effects has become more and more evident. The aims of ICP Materials are now to

- perform a quantitative evaluation of the effects of multi-pollutants such as S and N compounds, ozone and particles as well as climate parameters on the atmospheric corrosion of important materials, including materials used in objects of cultural heritage;

- describe and evaluate long-term corrosion trends attributable to atmospheric pollution in order to elucidate the environmental effects of pollutant reductions achieved under the Convention and in order to identify extraordinary environmental changes that result in unpredicted materials damage;

- use the results for mapping areas with increased risk of corrosion, and for calculation of cost of damage caused by deterioration of materials.

The present report addresses the second of these aims and gives the results of corrosion from the most recent (2008-2009) exposure programme for trend analysis. Complete reporting of environmental data from this exposure and, as a consequence, an analysis of trends in corrosion, soiling and pollution and their relation, will be reported in 2011. However, an overview of reported environmental data so far is included as well as quantified trends in key selected parameters based on the Guidelines on Reporting of Monitoring and Modelling of Air Pollution Effects.

2 Corrosion

2.1 Carbon steel

Carbon steel was prepared and evaluated by the sub-centre in Czech Republic (SVUOM).

Plates of unalloyed carbon steel (with C < 0.2 %, P < 0.07 %, Cr < 0.07 % according to CSN 11373) with dimensions 100 x 150 x 0,5 mm were used, triplicate per exposure period.

Before the exposure the samples were degreased in alkaline degreaser, rinsed with ethanol, dried and weighed.

The basic information about corrosion behaviour of carbon steel in different environmental conditions gives the value of mass change and corrosion loss. After exposure samples were weighed (conditioned in laboratory conditions) and the corrosion products were removed using the pickling method according to ISO 8407 (Annex A - solution C3.5.). In consecutive pickling cycles the corrosion layer was completely removed without dissolving the base metal. The repetition of the procedure stopped if the surface appears clean. The corrosion losses had been obtained by gravimetric method.

Corrosion losses for individual samples of carbon steel exposed in open atmosphere for 1 year exposure (2008/09) are summarised in Table 1 and presented in Figure 1. The test sites with the greatest corrosion losses of samples were No 3 (Kopisty), No 10 (Bottrop) and No 50 (Katowice) where industrial pollution still exists. The corrosivity is in category C2 for carbon steel only with exception above mentioned three test site where achieve category C3.

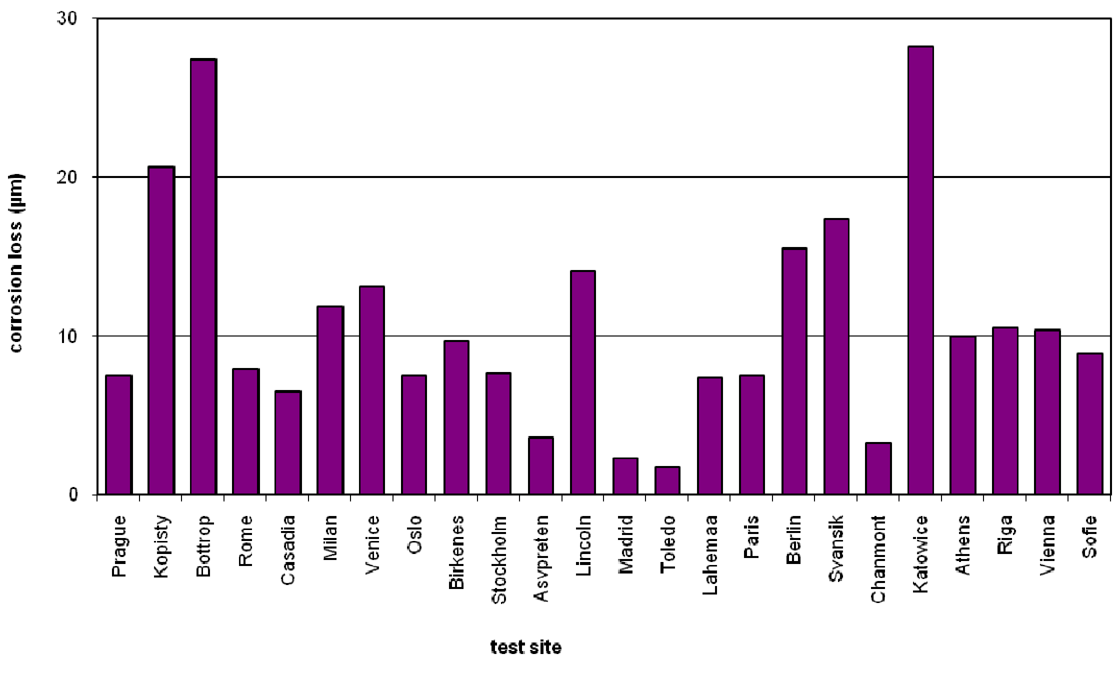

Figure 1 – Corrosion loss of carbon steel after 1 year exposure (2008-2009)

Table 1 - Corrosion losses (g/m^2) of carbon steel after 1 year exposure in open atmosphere (2008-2009)

Test site	ML_{91} (g.m^{-2})	ML_{92} (g.m^{-2})	ML_{93} (g.m^{-2})	ML (g.m^{-2})	ML (µm)
1 Prague	70,09	40,80	67,45	59,45	7,54
3 Kopisty	168,48	160,79	159,99	163,09	20,67
10 Bottrop	216,27	216,89	216,09	216,42	27,43
13 Rome	58,27	64,43	64,53	62,41	7,91
14 Cassacia	49,60	50,44	53,66	51,23	6,49
15 Milan	94,27	89,98	96,62	93,63	11,87
16 Venice	104,93	103,95	102,10	103,66	13,14
21 Oslo	58,01	58,62	60,99	59,21	7,50
23 Birkenes	78,89	73,38	77,39	76,55	9,70
24 Stockholm South	60,21	61,34	60,05	60,53	7,89
26 Aspvreten	26,80	28,07	30,07	28,31	3,59
27 Lincoln Cathedral	113,90	108,13	111,02	111,02	14,07
31 Madrid	18,38	18,63	17,65	18,22	2,31
33 Toledo	13,20	14,11	13,87	13,73	1,74
35 Lahemaa	56,75	58,69	58,71	58,05	7,36
40 Paris	58,92	58,02	61,11	59,35	7,52
41 Berlin	116,57	126,77	123,38	122,24	15,49
44 Svanvik	136,00	138,69	135,47	136,72	17,33
45 Chaumont	26,22	25,48	25,88	25,86	3,28
50 Katowice	220,84	223,98	223,06	222,63	28,22
51 Athens	78,76	76,58	80,71	78,68	9,97
52 Riga	81,03	84,04	83,16	82,74	10,49
53 Vienna	83,11	82,91	78,69	81,57	10,34
54 Sofia	60,99	74,15	75,26	70,13	8,89

2.2 Zinc

In the 2008-2009 exposure two types of zinc were exposed, one from the sub-centre in Switzerland (EMPA) and with a surface preparation resulting in a lightly higher corrosion attack and one from the sub-centre in Czech Republic (SVUOM).

2.2.1 Zinc from the sub-centre in Switzerland

Plates of zinc (99.99%) with dimensions of 100 x 2 mm were used, triplicate per exposure period. Before exposure the samples were degreased with petrol ether, glass blasted resulting in a rough surface (R_a about 2.9 µm). Pickling after exposure was done with concentrated glycine solution.

Corrosion losses for individual samples of zinc exposed in open atmosphere for 1 year exposure (2008/09) are summarised in Table 2. The highest values are observed in Kopisty, Bottrop, Birkenes, Aspvreten and Katowice. Birkenes and Aspvreten, rural sites in Norway and Sweden, are exceptions when comparing to the carbon steel results and points to a confounding factor specific for zinc. Present data points in the direction of organic acids but this will be discussed further in the 2011 report.

Table 2 Mass loss unsheltered zinc samples (1 year exposure) [g m^{-2}]

Site	Site No.	m_1-$m_1$ap	m_2-$m_2$ap	m_3-$m_3$ap	mean	standard deviation	Rel. standard deviation
		g m^{-2}	g m^{-2}	g m^{-2}	g m^{-2}	g m^{-2}	%
Prague	1	6.00	5.68	5.80	5.82	0.16	2.78
Kopisty	3	9.78	9.65	9.67	9.70	0.07	0.72
Bottrop	10	9.92	10.68	11.00	10.54	0.55	5.26
Rome	13	6.49	6.27	6.98	6.58	0.36	5.47
Casaccia	14	4.08	4.01	4.43	4.18	0.22	5.39
Milan	15	7.37	7.53	7.53	7.48	0.10	1.29
Venice	16	6.62	7.10	7.89	7.20	0.64	8.88
Oslo	21	6.81	6.32	6.26	6.46	0.30	4.67
Birkenes	23	10.45	10.55	10.77	10.59	0.16	1.53
Stockholm South	24	4.51	4.74	5.24	4.83	0.38	7.77
Aspvreten	26	9.87	10.15	11.72	10.58	1.00	9.41
Lincoln Cathedral	27	5.14	5.25	5.04	5.15	0.11	2.04
Madrid	31	4.28	4.50	5.08	4.62	0.41	8.96
Toledo	33	4.72	4.85	5.25	4.94	0.28	5.60
Lahemaa	35	6.58	6.50	8.39	7.15	1.07	14.91
Paris	40	4.64	4.31	4.49	4.48	0.17	3.76
Berlin	41	4.94	5.01	6.00	5.32	0.59	11.15
Svanvik	44	7.07	6.76	7.53	7.12	0.39	5.47
Chaumont	45	5.13	5.38	5.71	5.41	0.29	5.38
Katowice	50	9.61	9.39	10.13	9.71	0.38	3.91
Athens	51	5.29	4.80	5.35	5.15	0.30	5.92
Riga	52	4.96	5.34	6.83	5.71	0.99	17.36
Vienna	53	7.94	7.69	8.16	7.93	0.24	2.99
Sofia	54	3.48	3.54	3.90	3.64	0.23	6.21

m_i: mass before exposure, $m_i$ap: mass after exposure and pickling (i=1,2,3)

2.2.2 Zinc from the sub-centre in Czech Republic

Plates of zinc 98.5 % with dimensions 100 x 150 x 0,5 mm (rolled sheet) were used, triplicate per exposure period. Before the exposure the samples were degreased in acetone, dried and weighed.

The basic information about corrosion behaviour of zinc in different environmental conditions gives the value of mass change and corrosion loss. After exposure samples were weighed (conditioned in laboratory conditions) and the corrosion products were removed using pickling method according to ISO 9226 by glycine solution. In consecutive pickling cycles the corrosion layer was completely removed without dissolving the base metal. The repetition of the procedure stopped if the surface appears clean. The corrosion losses had been obtained by gravimetric method.

Corrosion losses for individual samples of zinc exposed in open atmosphere for 1 year exposure (2008/09) are summarised in Table 3. The test sites with the greatest corrosion losses of samples were No 3 (Kopisty), No 10 (Bottrop) and No 50 (Katowice) where industrial pollution still exists. The corrosivity is in category C2 for zinc only with exception above mentioned three test site where achieve category C3. Unfortunately, the sites Birkenes and Aspvreten were not included in this exposure on selected sites.

Table 3 - Corrosion losses (g/m^2) of zinc B after 1 year exposure in open atmosphere

Test site	ML_{91} ($g.m^{-2}$)	ML_{92} ($g.m^{-2}$)	ML_{93} ($g.m^{-2}$)	ML ($g.m^{-2}$)	ML (µm)
1 Prague	2,7433	3,1900	3,2967	3,0767	0,43
3 Kopisty	7,0733	7,0433	6,5567	6,8911	0,97
10 Bottrop	8,2600	7,1133	5,5700	6,9811	0,98
21 Oslo	1,1233	0,9600	0,0933	0,7256	0,10
27 Lincoln Cathedral	2,0033	2,3333	2,6300	2,3222	0,33
31 Madrid	1,2167	1,5700	1,2133	1,3333	0,19
35 Lahemaa	1,7433	2,9433	2,3967	2,3611	0,33
40 Paris	2,0667	1,8000	2,2633	2,0433	0,29
41 Berlin	3,5800	0,9967	1,5800	2,0522	0,29
45 Chaumont	1,9400	1,6467	2,0600	1,8822	0,26
50 Katowice	6,9667	5,8100	6,2067	6,3278	0,89
51 Athens	2,5867	1,7300	2,6067	2,3078	0,32

2.3 Limestone

Limestone was prepared and evaluated by the sub-centre in UK (BRE).

Surface recession at individual sites of limestone exposed in open atmosphere for 1 year exposure are summarised in Table 4. High recession rates are observed at several sites. One of the concerns with the 2005-2006 exposure were the unusually high observed recession rates at almost all sites and the suspicion that this could be related to changes in stone quality. Therefore additional exposures were performed on selected sites and with stones from the same quality as those exposed previously. The data from this exposure is shown in Table 5. The results show that there is some variation between the weathering rates of the different samples and that the variation does not seem to be reflected in the porosity. However, this variation could not explain all of the increase in the 2005-2006 values.

Table 4: Surface recession data in µm for Portland limestone exposed in unsheltered conditions showing the comparison with the earlier trend data.

Country	Name	1987-88 µm	1997-98 µm	2002-03 µm	2005-06 µm	2008-09 µm
		Unsheltered	Unsheltered	Unsheltered	Unsheltered	Unsheltered
01 Czech Rep	Prague-Bechovice	-22.90	-6.00	-5.33	-7.75	-4.50
03 Czech Rep	Kopisty	-19.40	-7.20	-7.91	-11.80	-11.11
05 Finland	Ähtäri	-15.00	-4.80	-5.46		
07 Germany	Waldhof Langenbrüge	-9.10	-3.30	-7.00		
08 Germany	Aschaffenburg	-8.50	-6.30	-7.62		
10 Germany	Bottrop	-17.40	-11.70	-11.88	-15.2	-8.56
13 Italy	Rome	-10.90	-6.00	-6.37	-8.87	-5.44
14 Italy	Casaccia	-6.70	-6.00	-6.71	-9.85	-5.93
15 Italy	Milan	-19.50	-9.00	-8.74	-15.09	-11.63
16 Italy	Venice	-8.80	-9.60	-7.17	-9.58	-6.98
21 Norway	Oslo	-11.70	-8.40	-7.31	-10.67	-4.32
23 Norway	Birkenes	-17.00	-7.50	-7.66	-14.65	-11.91
24 Sweden	Stockholm South	-14.00	-8.10	-7.29	-12.78	-9.57
26 Sweden	Aspvreten	-6.80	-3.60	-4.30	-7.79	-6.87
27 United Kingdom	Lincoln Cathedral	-9.90	-9.90	-7.63		-6.78
31 Spain	Madrid	-13.30	-3.30	-3.91	-8.11	-2.64
33 Spain	Toledo	-6.20	-7.50	-5.82	-11.48	-7.49
34 Russia	Moscow	-9.40	-9.00	-7.94		
35 Estonia	Lahemaa	-9.90	-5.40	-8.91	-10.65	-8.33
36 Portugal	Lisbon	-6.90	-11.40	-10.59		
37 Canada	Dorset	-8.90	-6.30		-14.31	
40 France	Paris	N/A	-8.70		-10.93	-8.14
41 Germany	Berlin	N/A	-3.30	-2.00	-8.59	-2.23
44 (73&76) Norway	Svanvik	N/A	-6.60	-9.11	-9.79	-12.79
45 Switzerland	Chaumont	N/A	-9.30	-9.04	-14.30	-7.06
46 United Kingdom	London	N/A	-7.50	-9.03		
50 Poland	Katowice					
51 Greece	Athens	N/A				-8.65
52 Latvia	Riga					-12.34
53 Austria	Vienna					-14.20
54 Bulgaria	Sophia					-5.35

Table: 5 Comparison of surface recession (default), porosity and saturation coefficient data from the different samples of Portland Whit bed exposed over the period 1987- 2009.

Country	**Site**	**1987 Stone**	**1997-2003**	**2005 Stone**	**2008 Stone**	**Ratio**
Czech Republic	1. Prague	-4.73			-4.50	0.95
Norway	23. Birkenes	-14.78			-11.91	0.81
United Kingdom	27. Lincoln	-5.05	-4.55	-5.89	-6.78	1.34
	Porosity	22.40	16.20	18.85	23.60	
	Saturation coeff.	0.58	0.72	0.62	0.64	
Switzerland	45. Chaumont	-7.06			-8.18	1.16
Poland	50. Krakow	-7.20			-11.25	1.56
Greece	51. Athens	-6.68			-8.65	1.29

3 Soiling

3.1 Modern glass

Modern glass was prepared and evaluated by the sub-centre in France (LISA).

Parameters for glass soiling for individual samples of modern glass exposed in sheltered atmosphere for 1 year exposure (2008/09) are summarised in Table 6 and figure 2 presents the haze data, which is the most important parameter. The highest levels are observed at the Athens test site.

Table 6. Detail of modern glass exposure (number and name of exposure site, beginning, end and duration of exposure period) and soiling analysis (TP/S in µg.cm^{-2}: total deposited particles per surface unit and Haze in %).

Exposure Site	Exposure time			TP/S µg.cm^{-2}	Haze %	ΔR= R_0-R %
	Start date	End date	Duration (days)			
Prague Lethany 1	23/10/2008	22/10/2009	364	2.60	1.68	1.12
Kopisty 3	24/10/2008	20/10/2009	361	broken	2.47	1.60
Bottrop 10	07/10/2008	07/10/2009	365	broken	2.00	1.72
Rome 13	29/10/2008	04/11/2009	371	7.00	3.37	1.44
Casaccia 14	14/10/2008	02/11/2009	384	49.70	3.24	1.88
Milan 15	21/10/2008	28/10/2009	372	broken	2.78	2.73
Venice 16	23/10/2008	27/10/2009	369	15.30	3.69	2.93
Oslo 21	07/10/2008	05/10/2009	363	2.70	1.61	1.09
Birkenes 23	08/10/2008	06/10/2009	363	broken	1.29	0.67
Stockholm 24	02/10/2008	14/10/2009	377	broken	3.44	1.88
Aspvreten 26	30/09/2008	01/10/2009	366	6.00	1.60	0.83
Lincoln Cath. 27	29/10/2008	30/11/2009	397	6.00	1.84	1.56
Madrid 31	02/10/2008	01/10/2009	364	broken	2.54	1.7
Toledo 33	01/10/2008	05/10/2009	369	4.00	2.95	1.23
Lahemaa 35	09/10/2008	06/10/2009	362	broken	1.58	1.12
Paris 40	03/10/2008	01/10/2009	363	6.30	2.88	1.79
Berlin 41	14/10/2008	22/10/2009	373	27.30	4.56	3.43
Svanvik 44	14/10/2008	08/10/2009	359	broken	1.51	0.89
Chaumont 45	02/10/2008	01/10/2009	364	0.70	1.02	0.63
Katowice 50	07/10/2008	15/10/2009	373	28.00	2.97	2.48
Athens 51	01/10/2008	01/10/2009	365	63.30	10.48	6.1
Riga 52*	11/11/2008	13/11/2009	367	1181.70*	54.78*	54.47
Vienna 53	10/10/2008	12/10/2009	367	2.30	2.05	1.24
Sophia 54	10/10/2008	10/10/2009	365	broken	2.96	1.73

*: sample exposed in horizontal position

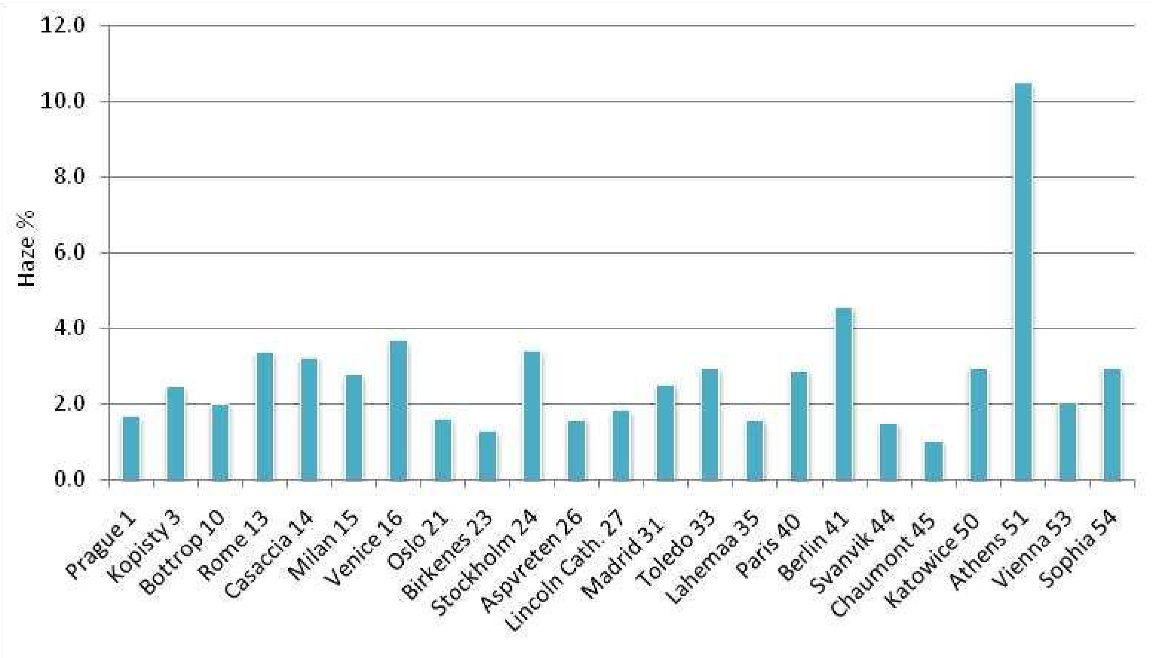

Figure 2: %Haze measured on modern glass samples exposed 1 year in sheltered condition.

3.2 Teflon filter

The main research centre in Sweden (Swerea KIMAB) was responsible for the exposure and evaluation of the soiling of the Teflon filter.

Parameters for loss of reflectance for individual samples of Teflon filter exposed in sheltered atmosphere for 1 year exposure (2008/09) are summarised in Table 7. The highest values are observed at the Berlin and Athens sites.

Table 7. Detail of Teflon filter exposure (name of exposure site, beginning, end and duration of exposure period) and soiling analysis (loss of reflectance).

Exposure Site	Exposure time			$Ln(R_0/R)$
	Start date	End date	Duration (days)	
Prague	2008-10-23	2009-10-20	363	0,005
Kopisty	2008-10-24	2009-10-20	362	0,013
Bottrop	2008-10-07	2009-10-07	365	0,006
Rome	2008-10-29	2009-11-04	371	0,020
Casaccia	2008-10-17	2009-11-02	381	0,005
Milan	2008-10-21	2009-10-29	373	0,030
Venice	2008-10-23	2009-10-27	369	0,012
Berlin	2008-10-14	2009-10-22	373	0,069
Oslo	2008-10-07	2009-10-05	363	0,019
Birkenes	2008-10-08	2009-10-06	363	< D.L.
Svanvik	2008-10-14	2009-10-06	357	< D.L.
Stockholm	2008-10-02	2009-10-14	377	0,010
Aspvreten	2008-09-30	2009-10-01	366	0,007
Lincoln	2008-10-29	2009-11-30	397	0,022
Madrid	2008-10-02	2009-10-01	364	0,018
Toledo	2008-10-01	2009-10-05	369	< D.L.
Lahemaa	2008-10-09	2009-10-06	362	0,011
Paris	2008-10-03	2009-10-01	363	0,017
Chaumont	2008-10-02	2009-10-01	364	0,015
Katowice	2008-10-07	2009-10-15	373	0,038
Athens	2008-10-01	2009-10-01	365	0,056
Riga	2008-11-11	2009-11-13	367	0,046
Vienna	2008-10-10	2009-10-12	367	0,007

4 Quantified trends in selected parameters based on the Guidelines on Reporting of Monitoring and Modelling of Air Pollution Effects.

ICP Materials exposes materials and measures key parameters every three years in the trend exposures. The results prior to the 2008–2009 exposure have already been reported (see ECE/EB.AIR/WG.1/2008/3). The results indicated a significant corrosion decrease at ICP Materials test sites. The first soiling and corrosion results from the period 2008–2009 have become available and are presented in this report. They have enabled the updating of trends for carbon steel, one of the key parameters in annex II of the Guidelines on reporting of monitoring and modelling of air pollution effects (see figure 1). Additional actions are needed in order to meet the 2020 and 2050 targets as there were no substantial changes in corrosion when comparing the exposures in 2005–2006 and 2008–2009.

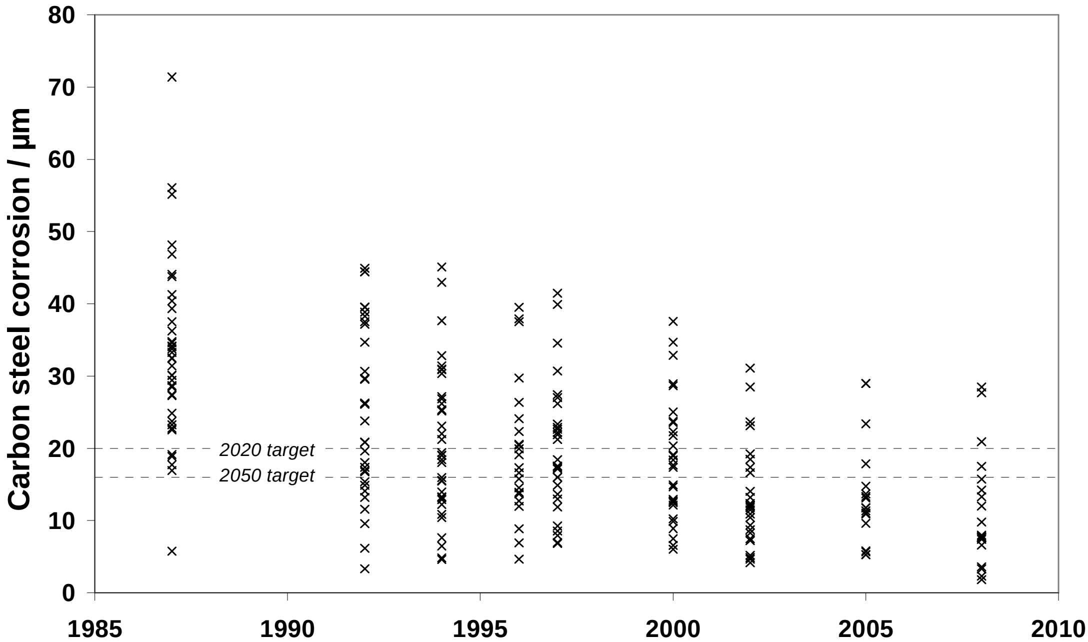

Figure 3. Carbon steel corrosion in the period1987-2008 measured at ICP Materials test sites. Dashed lines indicate targets for protecting infrastructure and cultural heritage for 2020 and 2050 (see ECE/EB.AIR/WG.1/2009/16).

Regarding soiling, the present exposure (2008-2009) was the second one-year exposure of soiling materials. The first was in the period 2005-2006, and recent data enabled first evaluation of soiling trends. For haze (Figure 4), which is the most suitable parameter for evaluation of soiling, 23 per cent of the sites had an increasing trend, 41 per cent no trend and 36 per cent a decreasing trend.

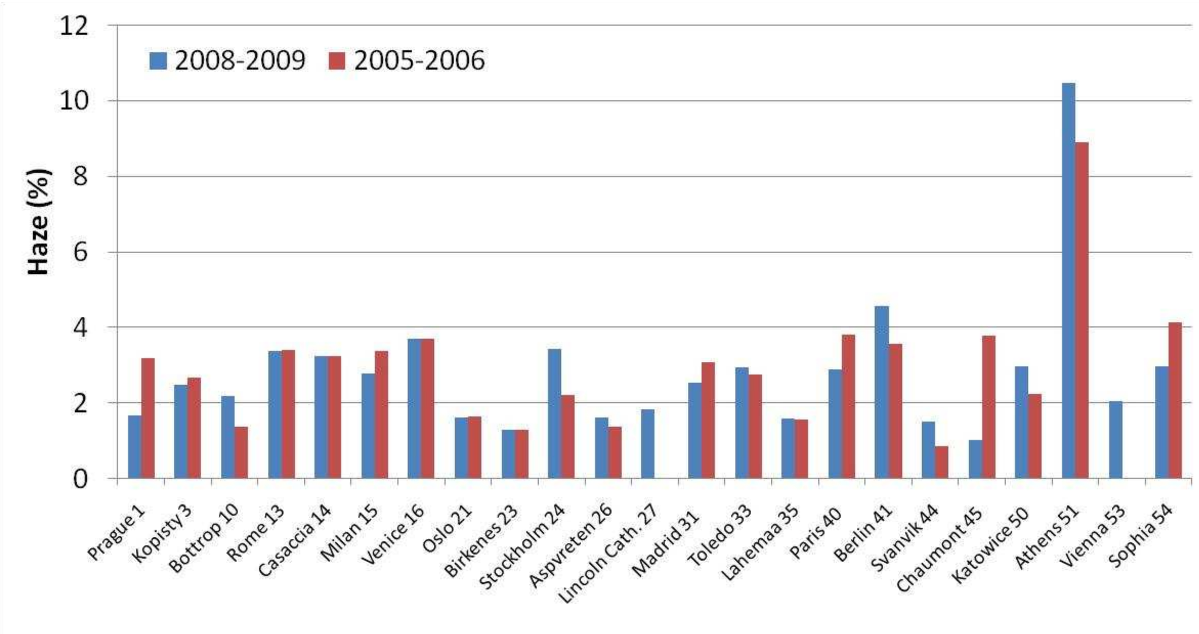

Figure 4 Trends in haze measured on modern glass samples exposed 1 year in sheltered condition.

5 Overview of reported environmental data

Reporting of environmental data is usually a slower process than evaluation of corrosion attack since some of the data is frequently not measured directly by the members of ICP Materials. Instead, data are collected from different agencies with individual policies on release of data depending on validation procedures. The time lag is usually from ½ to one year. However, a large portion of the data has already been reported as presented in tables and 9. As is seen from table 8 there is a large portion of missing data regarding precipitation analysis but this data is expected to be completed in next years report.

Table 8: Available data from the ECE - ICP trend exposure programme 2008 – 2009, as percentage available data for 13 months from October 2008 to October 2009 (as reported by 2010-06-04).

	Mandatory (gases could also be measured by passive sampling, see table 9)											Optional	
	Climate		Gases (concentration)				Precipitation					Prec.	Particles
Station no	Temp	RH	SO_2	NO_2	O_3	HNO_3	Amount	H^+	SO_4^{2-}	NO_3^-	Cl^-	Cond.	PM10 (conc.)
	Availability (%)												
01	100	100	100	100	92	0	100	100	100	100	100	100	92
03	100	100	100	100	100	0	100	100	100	100	100	100	92
10	93	93	90	81	90	0	100	100	100	100	100	100	95
13	0	0	0	83	83	0	0	0	0	0	0	0	0
14	100	100	100	100	100	0	100	0	0	0	0	0	0
15	96	96	85	94	96	0	97	0	0	0	0	0	0
16	100	100	100	94	100	0	100	0	0	0	0	0	0
21	100	100	85	85	0	0	92	92	92	92	92	92	0
23	99	94	97	98	97	95	100	99	100	100	100	99	83
24	92	92	0	0	0	0	92	46	46	46	46	0	0
26	0	0	0	0	0	0	0	0	0	0	0	0	0
27	0	0	0	0	0	0	0	0	0	0	0	0	0
31	92	92	91	90	89	0	92	85	77	77	77	85	66
33	91	91	92	92	92	0	92	87	75	74	74	86	87
35	96	96	94	96	97	0	40	31	40	40	40	31	100
40	100	100	90	90	90	0	100	100	100	100	100	100	95
41	100	100	97	98	97	0	0	0	28	28	28	0	98
44	100	100	99	92	0	0	100	96	97	97	97	96	0
45	92	92	89	90	90	0	92	91	89	89	89	91	92
50	100	100	100	100	100	0	0	0	0	0	0	0	86
51	99	92	100	93	99	0	100	0	0	0	0	0	95
52	100	100	96	98	92	0	100	0	0	0	0	0	0
53	0	0	99	99	98	0	0	0	0	0	0	0	100
54	92	92	88	88	86	0	92	92	92	92	0	0	87

Table 9: Availability (%) of data from IVL measurements for the ECE - ICP trend exposure programme. October 2008 to October 2009.
Bi- or tri- monthly values.

		Particles	Temp	Gases (concentration)			
Station no	Station	Deposited mass (total and for separate ions)		HNO_3	SO_2	NO_2	O_3
		availability (%)					
1	Prague	75	100	100	0	0	0
3	Kopisty	50	75	100	0	0	0
10	Bottrop	100	100	100	0	0	0
13	Rome	100	100	100	0	0	0
14	Casaccia	100	100	100	0	0	0
15	Milan	100	100	100	0	0	0
16	Venice	100	100	100	0	0	0
21	Oslo	100	100	100	0	0	100
23	Birkenes	83	83	83	0	0	0
24	Stockholm, Södermalm	100	100	100	100	100	100
26	Aspvreten	100	100	100	100	100	100
27	Lincoln Cathedral	67	100	100	100	100	100
31	Madrid	100	100	100	0	0	0
33	Toledo	100	100	100	0	0	0
35	Lahemaa	100	100	100	0	0	0
40	Paris	67	100	100	0	0	0
41	Berlin	100	100	100	0	0	0
44	Svanvik	83	100	100	0	0	100
45	Chaumont	100	50	100	0	0	0
50	Katowice	100	100	100	0	0	0
51	Athens	100	100	0	0	0	0
52	Riga	100	100	83	83	83	83
53	Vienna	100	100	100	0	0	0
54	Sofia	0	0	0	0	0	0

6 Conclusions

- ICP Materials has completed the 2008-2009 trend exposure resulting in corrosion data on carbon steel, zinc and limestone and soiling data on modern glass and Teflon filter for the network of test sites. Reporting of environmental data are proceeding according to plans and will be finalised in 2011.
- The soiling and corrosion results from the period 2008–2009 have enabled the updating of trends, which will be further elaborated in the 2011 report when comparison with environmental data is possible. Additional actions are needed in order to meet the 2020 and 2050 targets for carbon steel as there were no substantial decreases in corrosion when comparing the exposures in 2005–2006 and 2008–2009.

7 References

1. http://www.unece.org/env/lrtap/WorkingGroups/wge/welcome.html